

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

Course No: MTE 3102

Course Title: Control Systems Sessional

Experiment No: 05

Experiment Name: Design of a LQR controller for a given system using trial & error method

Submitted To:

Md. Faisal Rahman Badal

Assistant Professor

Department of Mechatronics Engineering
Rajshahi University of Engineering & Technology

Safwat Mukarrama Choudhury

Lecturer

Department of Mechatronics Engineering,
Rajshahi University of Engineering & Technology

Submitted by:

Adiba Binta Newaz
2108043

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Experiment No: 5

Experiment name: Design of a LQR controller for a given system using trial & error method.

Objectives:

1. To design a Linear Quadratic Regulator (LQR) controller for a given linear time-invariant system.
2. To apply the trial-and-error method for tuning Q and R to achieve desired dynamic response characteristics (e.g., stability, minimal overshoot, and fast settling time).
3. To analyze the performance of the closed-loop system under different weight configurations.

Theory:

Linear Quadratic Regulator (LQR) is a mathematical method used in optimal control theory to design a controller that regulates the state of a linear dynamic system while minimizing a cost function. LQR is widely used in automatic control systems due to its ability to achieve optimal performance with minimal control effort.

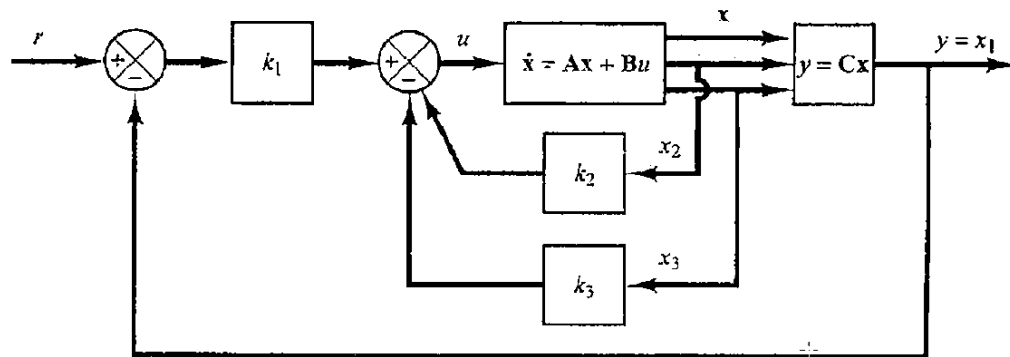


Figure 1: Block Diagram of a basic LQR system [1]

Key Features of LQR:

- It is based on state-space representation.
- Provides optimal control by minimizing a quadratic cost function.
- Balances performance (e.g., fast response) and control effort (e.g., energy consumption).
- The solution involves solving an Algebraic Riccati Equation (ARE)

Cost Function Criteria:

The LQR approach allows multiple criteria for optimizing the performance index:

1. **Minimum Time:** Time taken to reach steady state (90% final value)
 - Performance Index: $J = T$, where T is the final time and J is Performance criterion/ Cost function.
2. **Minimum Final Error:** Achieve the desired final state with minimal error.
 - Cost function: $J = \|x(t)\|$; $x(t)$ = System Trajectory

3. **Minimum Transient Response:** Improve the transient characteristics (e.g., overshoot, settling time) by minimizing area under $\|x(t)\|^2$.
 - Cost function: $J = \int_0^T \|x(t)\|^2 dt$
4. **Minimum Control Input:** Minimize the control input $u(t)$
 - Cost function: $J = \int_0^T \|u(t)\|^2 dt$

Quadratic Cost Function [2]:

$$J = x^T(t)Sx(T) + \int_0^T [x^T(t)Qx(t) + u^T(t)Ru(t)]dt$$

Here, S = weighting matrix; symmetric, non-negative definite

Q = weighting matrix; symmetric, non-negative definite

R = weighting matrix; symmetric, positive definite

S, Q, R are quadratic regulator.

Designing LQR Controller:

Given a system in state-space form:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned}$$

Assuming disparity matrix, $D=0$, the goal is to determine the control input $u(t)$ such that a quadratic cost function is minimized:

$$J = \int_0^\infty [x^T(t)Qx(t) + u^T(t)Ru(t)]dt$$

The optimal control law is:

$$u(t) = -Kx(t)$$

Where the gain matrix K is:

$$K = R^{-1}B^TP$$

Here, P is the unique positive solution of the matrix

B is the control input co-efficient matrix

R is the control input weighting matrix

Algebraic Riccati Equation (ARE) [3]:

$$A^TP + PA - PBR^{-1}B^TP + Q = 0$$

Substituting the control law into the system yields the closed-loop dynamics:

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ &= Ax(t) + B(-Kx(t)) \\ &= Ax(t) - KBx(t) \\ &= [A - KB]x(t) \\ &= A_c x(t); \text{ where } [A - KB] = A_c \end{aligned}$$

Required Software:

- MATLAB

State Space Example:

Let, a sample state-space system is represented as:

$$\dot{x} = \begin{bmatrix} -1 & -5 \\ 10 & -1 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

Code:

```
clc;
clear all;
clf;
% Define State-Space Matrices
A = [-1 -5; 10 -1];
B = [10; 0];
C = [1 0];
D = 0;

% Create State-Space System
sys1 = ss(A, B, C, D);
Cr=ctrb(A,B);
Rank1=rank(Cr);
Rank2=rank(A);

if (rank(Cr)==rank(A))
disp("System 1 is Contrallable");
else
disp("System 1 is not Contrallable");
end;
Q=15*C'*C;
R=0.015;
K=lqr (A,B,Q,R);
Ac=A-B*K;
sys2=31.65*ss(Ac,B,C,D);
figure(1)
step
step(sys1)
hold on
step(sys2)
```

Result:

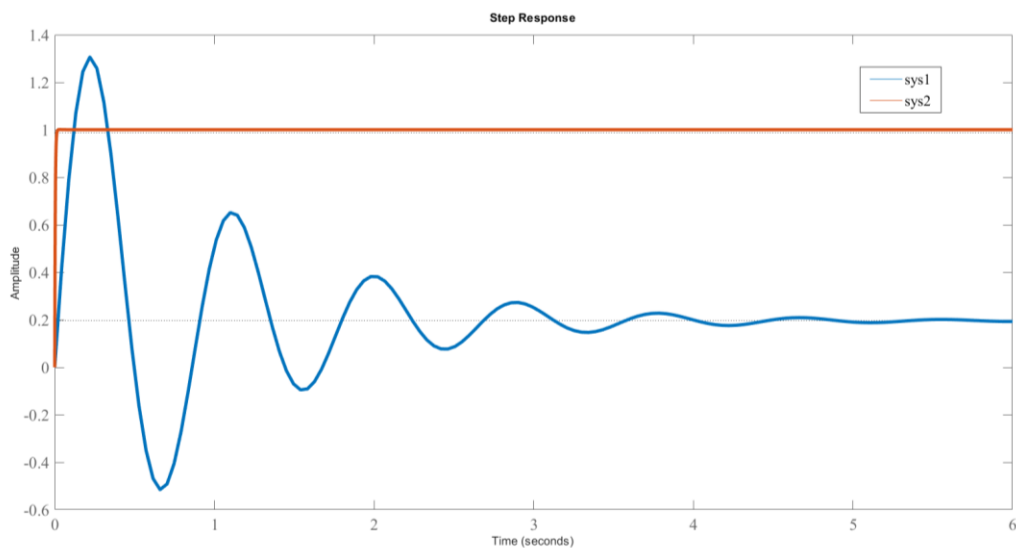


Figure 2: Output Performance of an LQR system

```

Command Window

System 1 is Contrallable
Here is an example of how the function step works:
Consider a randomly generated stable Transfer Function Model:
of the form  $G(s)=num(s)/den(s)$  :

num =

      0   -0.1022    0.0316    0.1934   -0.1795    0.1620

den =

    1.0000    6.2190   50.6538  222.7866  359.5180  162.7478

```

Figure 3: Controllability test of the system in MATLAB

Performance Analysis:

Controller	Rise Time	Settling Time	Peak Time	Overshoot (%)	Peak Value
System 1	0.0573	0.1215	0.22	5.66	1.31
System 2	0.0066	0.0108	0.0319	1.2996	1.0009

Discussion & Conclusion:

The performance analysis clearly shows that System 2 outperforms System 1 in all key time-domain metrics. With a significantly faster rise time (0.0066 s vs. 0.0573 s) and settling time (0.0108 s vs. 0.1215 s), System 2 responds more quickly and stabilizes faster. Additionally, it exhibits a much lower overshoot (1.30% compared to 5.66%), indicating better damping and control precision. The peak value of System 2 (1.0009) is nearly ideal, while System 1 overshoots to 1.31. These results highlight that the LQR controller used in System 2 was more effectively tuned, leading to superior performance. This demonstrates the importance of selecting appropriate weighting matrices in LQR design to achieve optimal system behavior with minimal control effort.

References:

- [1] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice Hall, 1990.
- [2] K. Ogata, *Modern Control Engineering*, 5th ed., Pearson Education, 2010.
- [3] R. C. Dorf and R. H. Bishop, *Modern Control Systems*, 13th ed., Pearson, 2017.